Predicting Mortality

Using Non-communicable Disease Risk Factors

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World Health NCD vs Mortality Rate (1980 - 2021)

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Task - Predict the mortality rate for 25-29 year olds using these predictors

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- Adult_curr_cig_smoking (current adult cigarette smoking)
- Adult_curr_tob_use (any current adult tobacco use)
- NCD_BMI_25A (prevalence of BMI ≥ 25, adults)
- NCD_BMI_30A (prevalence of BMI ≥ 30, adults)
- NCD_PAA (physical activity, adults)
- NCD_PAC (physical activity, children)
```

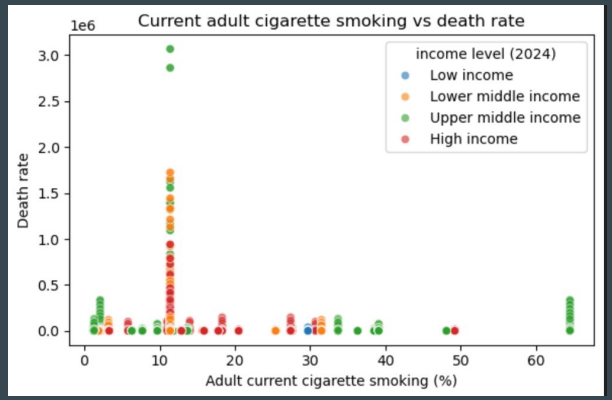
Model

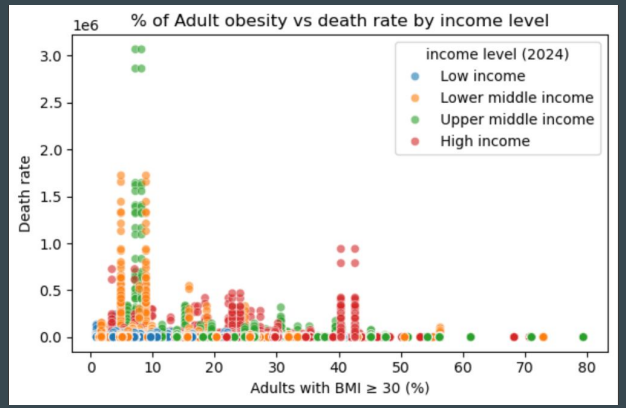
Linear Regression:
Test $R^2 = 0.03$ | Test RMSE = 2,278
Not able to capture complex relationships

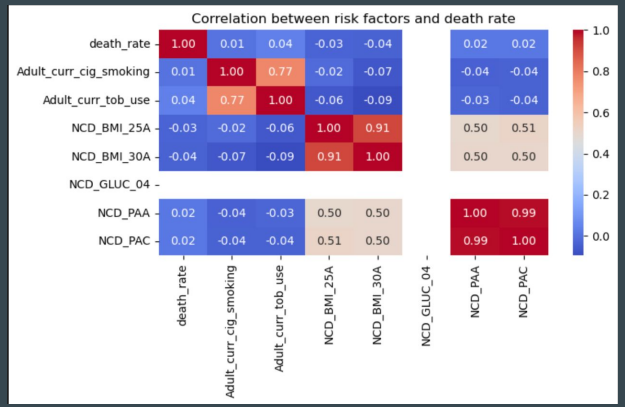
```
Validation RMSE: 4633.585479564703
R^2: -0.005833024647648566
Test RMSE: 2278.365070773903
Test R^2: 0.028333877968050647
```

Random Forest **(Selected)** :
Test $R^2 = 0.09$ | Test RMSE = 2,210
Captured some nonlinear patterns but only some variation

```
Validation RMSE: 4305.675074963619
Validation R^2: 0.13149156986393795
Test RMSE: 2210.0727272873073
Test R^2: 0.08571085458876893
```







Results & Limitations

- Low R^2 : NCD factors only explain about 9% of the death rate variation
- Unobserved factors like health systems, infectious disease, conflict, socioeconomic factors
- Could be useful for comparing countries mortality risk, not for precise decision making
- **Key Results:** Mortality has many different contributing factorsç